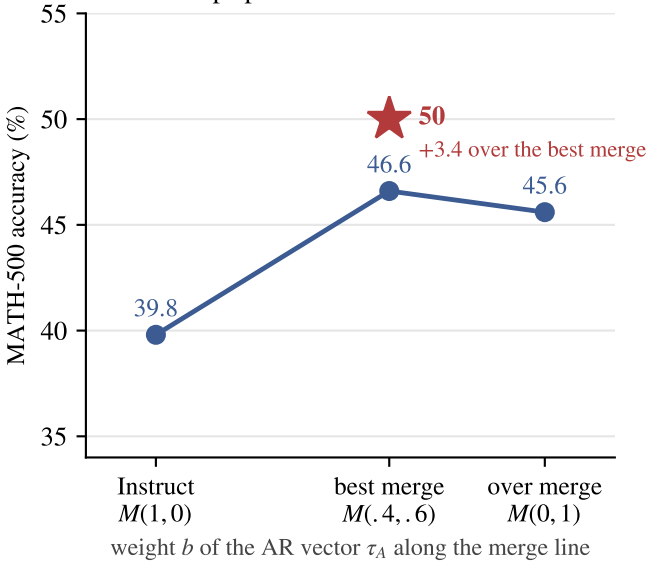
Over distillation: the over-merged model knows more per token but cannot finish — borrow its tokens only where the best merge is unsure

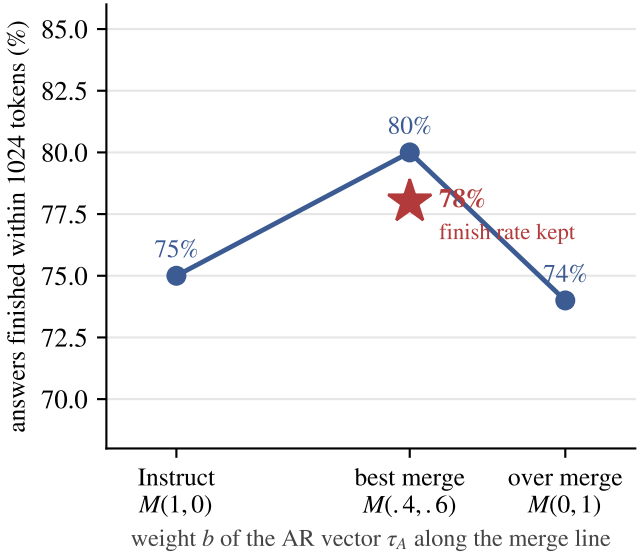
one model on the merge line (1× decode)

gate $M(.4,.6) \Leftarrow M(0,1)$ @ θ=0.8: best merge decodes, over merge supplies its unsure tokens (2× decode)

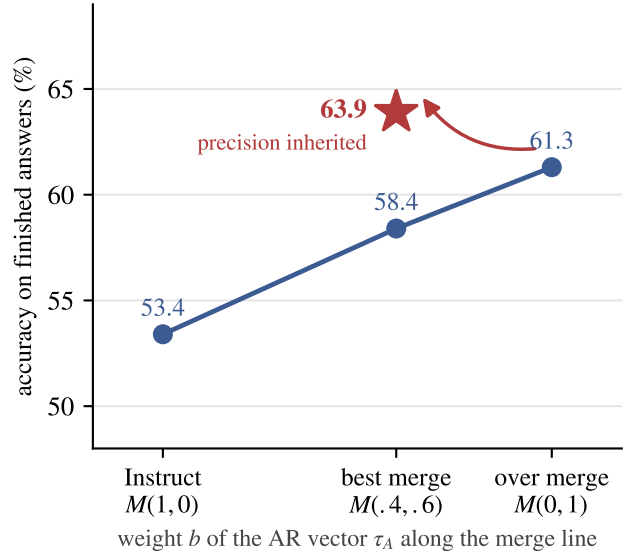
(a) Accuracy peaks at the best merge, and drops past it …



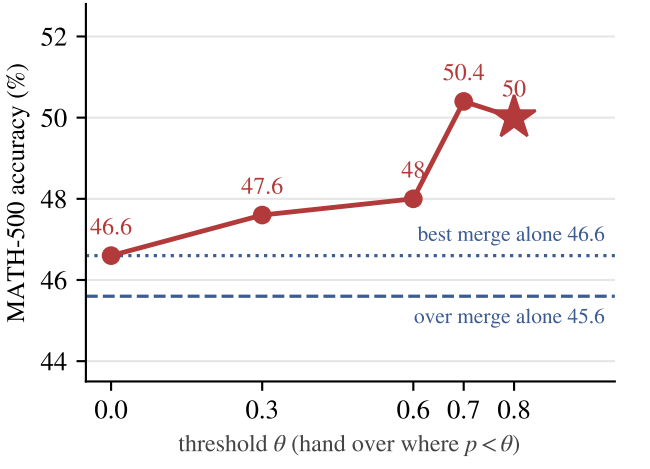
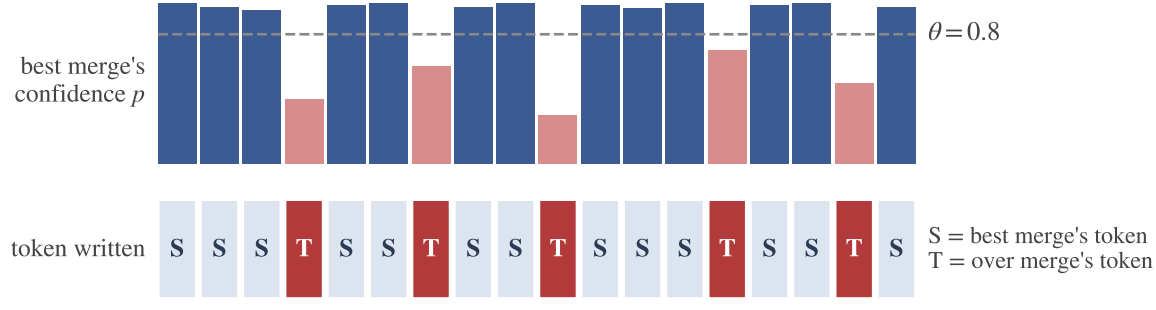
(b) … because the over merge stops finishing its answers …



(c) … although each finished answer is more often right.



(d) Threshold sweep: above both ends from $\theta = 0.3$, flat between 0.7 and 0.8



(e) The gate. The best merge decodes left to right and keeps its own token where it is confident ($p \geq \theta$); where it is unsure ($p < \theta$) the over merge's token is written instead. At $\theta = 0.8$ about 24% of the tokens are handed over and about 3% actually change — enough to inherit the precision without losing the finish rate.